RESEARCH PLAN

Integrated Doctoral Education Program

Name	Peeradon, Sarnkaew
Nationality	Thai
Institution	Department of Mathematics and Computer Science, Chulalongkorn University
Interesting Fields	Graph Neural Networks · Representation Learning · Dynamic Graphs
Proposed Research Theme	Heterophilic Heterogeneous Graph Neural Networks

1 Present Field of Study

During my undergraduate studies at Chulalongkorn University, I developed a strong interest in machine learning, progressing from classical methods (Gaussian mixture models for depression classification) through spatiotemporal deep learning (Conv-LSTM for drought forecasting), and ultimately to graph neural networks for relational data. My participation in the JAIST Internship Program during my third year further strengthened my commitment to a research career.

My current focus is **temporal representation learning for continuous-time dynamic graphs**, specifically how to model absolute and relative temporal signals jointly. My recent first-author paper at PAKDD 2026 (Data Science: Foundations and Applications) contributes three advances: (1) a *dual-stream temporal encoder* with proven time-rescaling invariance; (2) *K-MOTE*, a KAN-based mixture-of-time-experts integrating Fourier, B-spline, and shock-wavelet bases with continuous gating; and (3) a *temporal-modulated Mamba-2 fusion* module using FiLM conditioning within a continuous-time state-space formulation. These components address the challenge of learning expressive, generalizable time encoders for irregular interaction streams, and collectively demonstrate my ability to combine mathematical rigour with empirical effectiveness.

2 Research Proposal: Heterophilic Heterogeneous GNNs

2.1 Introduction and Research Questions

Real-world relational data, including citation networks, recommender systems, and knowledge bases, naturally involve *multiple node and edge types*, motivating heterogeneous information networks (HINs) [1] and heterogeneous graph neural networks (HGNNs) [2–5]. Yet heterogeneity alone does not capture the full difficulty: two orthogonal challenges remain largely unaddressed together. *Heterophily* (connected nodes having different labels) causes naive message-passing to degrade, and *temporal dynamics* (discrete- or continuous-time graph evolution) add a third dimension. Addressing all three jointly defines the scope of this five-year doctoral programme, structured around five research questions.

Q1 — Proper heterophily metrics. Existing heterophily measures in heterogeneous graph either project the graph to homogeneous meta-path views (discarding cross-type edges [6, 7]) or require a pre-trained model to generate pseudo-labels [8], making them unsuitable as model-

agnostic diagnostics. No existing measure satisfies all five axiomatic properties (Max, Min, ACB, ECT, Mon) [9] for the heterogeneous setting. We propose **five typed heterophily metrics** (TALH, TASH, TAFH, TACH, TLI) that operate directly on the full heterogeneous graph without projection or pre-training, each with formal axiomatic guarantees.

Q2 — Heterogeneity-aware representation for heterogeneous graphs. Several HGNNs such as HAN [10], HGT [11], and SeHGNN [3] use unconstrained type-specific projections with no guarantee that distinct types remain separable [12]. Geometric methods expose the same gap: KG embeddings (TransE, RotatE, SpherE, SKGE) use a single shared manifold, while HHNE [13] places all nodes in one hyperbolic space and HHNE++ partitions by *relation* rather than *node type*, providing no inter-type separation guarantee. HCAN [14] applies vMF priors but only for homogeneous networks. No method explicitly disentangles label-predictive from label-independent features within each type. We propose **STAGE**: type projections are constrained to the Stiefel manifold for unconditional inter-type angular separation, with vMF anchors initialised via the Tammes problem enforcing intra-type inter-class separation. Pre-trained once, STAGE serves as a plug-in initialisation for any downstream HGNN.

Q3 — Static heterophilic heterogeneous graphs. Under heterophily, standard message-passing causes over-smoothing, whereby nodes become indistinguishable, and this effect worsens in the typed setting. Existing heterophily-aware HGNNs (ADHGN, HDHGR, Hetero2Net, H2SGNN) [6, 15–17] all reduce the graph to meta-path-projected homogeneous views, discarding direct cross-type edges. No existing model simultaneously handles inter/intra-type heterophily, meta-path higher-order relations, and over-smoothing while keeping graph structure within a single framework. We propose **H^{2}S-NN**: the heterogeneous graph is *lifted* to a typed 2D cellular complex (nodes as 0-cells, edges as 1-cells, meta-path instances as 2-cells), equipped with a cellular sheaf whose relation-specific Hodge Laplacians prevent over-smoothing while preserving typed relational semantics and dissimilarity between nodes (heterophily).

Q4 — Discrete-time dynamic heterophilic heterogeneous graphs. Discrete-time dynamic heterogeneous graphs model graph evolution as a sequence of typed snapshots $G_1, G_2, \ldots, G_T$, where each snapshot captures the graph state within a time interval. All existing DT-DHGNNs [3, 18–21] inherit the heterophily assumption and conflate two fundamentally distinct neighbourhood signals: *persistent* (stable across snapshots) and *cumulative* (accumulating over time, analogous to sequential modelling in recurrent architectures). A recent category-theoretic result [22] proves that conflating them causes irreversible information loss. No method can detect or exploit heterophily drift across snapshots. We propose **DistH^{2}SNN**: the static H^{2}S-NN is extended to discrete-time dynamics by decomposing each snapshot's neighbourhoods into persistent and cumulative streams, tracking heterophily drift with a temporal gate, and fusing streams via dual Mamba encoders. This structure preserves heterophily, heterogeneity, and temporal dependencies in both persistent and cumulative aspects.

Q5 — Continuous-time dynamic heterophilic heterogeneous graphs. Unlike discrete-time models, continuous-time dynamic graphs represent graph evolution as a stream of timestamped events (u, r, v, t), where each interaction is processed at its exact arrival time without discretisation into snapshots. Real-world heterogeneous networks (recommender systems, financial transactions) grow as irregular event streams recorded in UNIX timestamps, which are

difficult to model as discrete snapshots. Recent continuous-time heterogeneous GNNs [23–26] do not account for heterophily; only isolated works address heterophily in homogeneous temporal graphs [27]. We propose **ContH^{2}S-NN**: H^{2}S-NN is scaled to continuous-time by augmenting node and edge embeddings with absolute time encodings, upgrading the 2D cellular sheaf with temporal edges, and using sheaf-energy-based ODE diffusion to preserve heterophily, heterogeneity, and multi-hop temporal dependencies.

Research thesis. Q1–Q5 converge on a unified thesis: *principled heterogeneous graph learning requires jointly axiomatising heterophily measurement, enforcing geometric type separation, and propagating signals through a topological structure that preserves typed relational semantics across static and dynamic settings.* Table 1 summarises this programme.

Research Question	Gap in Prior Work	Proposed Contribution
Q1. Proper heterophily metrics	Metrics distorted by meta-path projection or require a pre-trained model; no axiomatic grounding	Five heterophily similarity and label informativeness measures (TALH, TASH, TAFH, TACH, TLI) satisfying heterophily desired properties
Q2. Heterogeneity-aware representation for heterogeneous graphs	No formal inter/intra-type separation guarantee; no disentanglement of label-predictive from label-independent features	Stiefel-manifold encoder with provable angular inter-type (G1) and vMF intra-type class separation (G2)
Q3. Static heterophilic heterogeneous graphs	No HGNN jointly addresses heterophily, meta-paths, and over-smoothing	Sheaf diffusion on typed 2D cellular complex
Q4. Discrete-time dynamic heterophilic heterogeneous graphs	Conflate persistent/cumulative signals; no heterophily drift detection across snapshots	Extended from H^{2}S-NN Persistent–cumulative decomposition with temporal heterophily gate and dual Mamba streams
Q5. Continuous-time dynamic heterophilic heterogeneous graphs	Extended from H^{2}S-NNN, CT-HGNN accounts for heterophily; irregular event streams incompatible with snapshot-based modeling	Temporal complex with sheaf ODE diffusion over event streams, coupled with a non-dissipative global state ODE

Table 1: Five-component research programme: gaps and proposed contributions.

Alignment with Murata's lab. This proposal directly aligns with the lab's work on dynamic graphs [28] and heterogeneous graphs [29, 30]. I aim to develop methods grounded in algebra, geometry, and spectral topology, with emphasis on both mathematical rigour and empirical validation.

3 Proposed Framework

The five components form a unified pipeline, illustrated in Figure 1. Proposed heterophily metrics (TALH, TASH, TAFH, TACH, and TLI) measure heterophily in the dataset (Q1). STAGE pre-trains geometric embeddings (Q2); H^{2}S-NN performs static sheaf-diffusion classification (Q3); DistH^{2}SNN and ContH^{2}S-NN extend to temporal settings (Q4, Q5). Heterophily metrics (Q1) also provide diagnostic signals throughout, as shown in Figure 1.

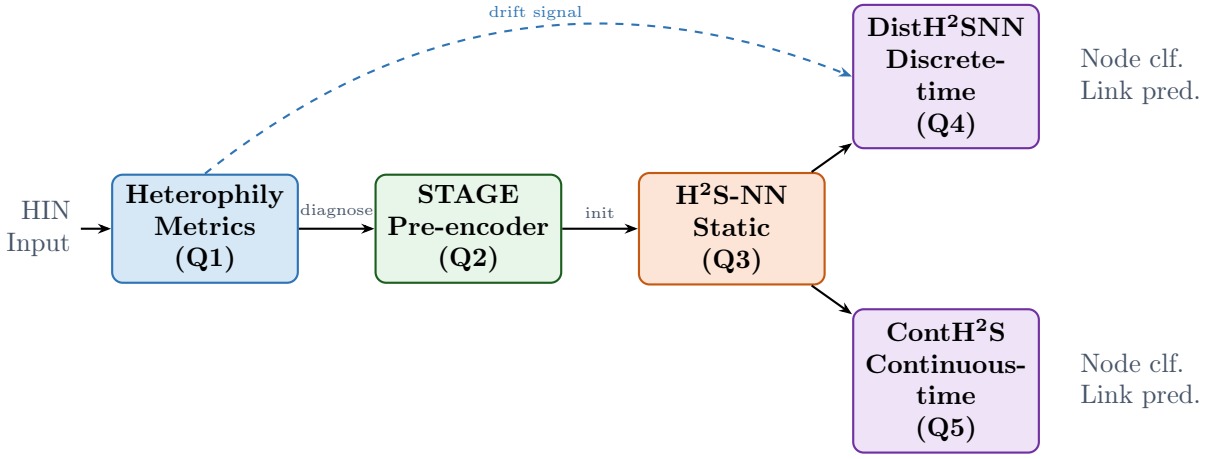


Figure 1: Unified five-component pipeline. Heterophily metrics (Q1) serve as diagnostic signals throughout. STAGE (Q2) provides heterogeneity-aware representation for all downstream components.

3.1 Component I (Q1): Typed Heterophily Metrics

Gap. Existing metrics project the heterogeneous graph to homogeneous meta-path views (discarding cross-type edges) or require a pre-trained model. No measure satisfies all five axiomatic properties for the heterogeneous setting.

Proposed Approach. For unlabelled nodes, a soft label q_u is propagated from labelled neighbours within 3 hops. Each node is characterised by three *atom distributions*: a label feature, a typed structural feature, and a label-independent feature (feature vectors with the label-predictive component removed). Agreement scores (A_r) are adjusted against a constant baseline (B_r), passed through a sign adjustment function (Ψ_r) calculated per relation, and finally yield signed scores in the range $[-1, 1]$, which instantiate into TALH (label), TASH (structural), TAFH (feature), and their summation TACH, each of which satisfies the five axiomatic properties (Section 5.5). TLI further distinguishes informative from non-informative heterophily via mutual information, with proven Max and ACB properties, as shown in Figure 2.

Key novel claim: To our knowledge, TALH, TASH, TAFH, and TACH are the first heterogeneous-graph metrics to jointly satisfy all five axiomatic properties (Max, Min, ACB, ECT, Mon), and TLI is the first label-informativeness measure defined for the heterogeneous setting.

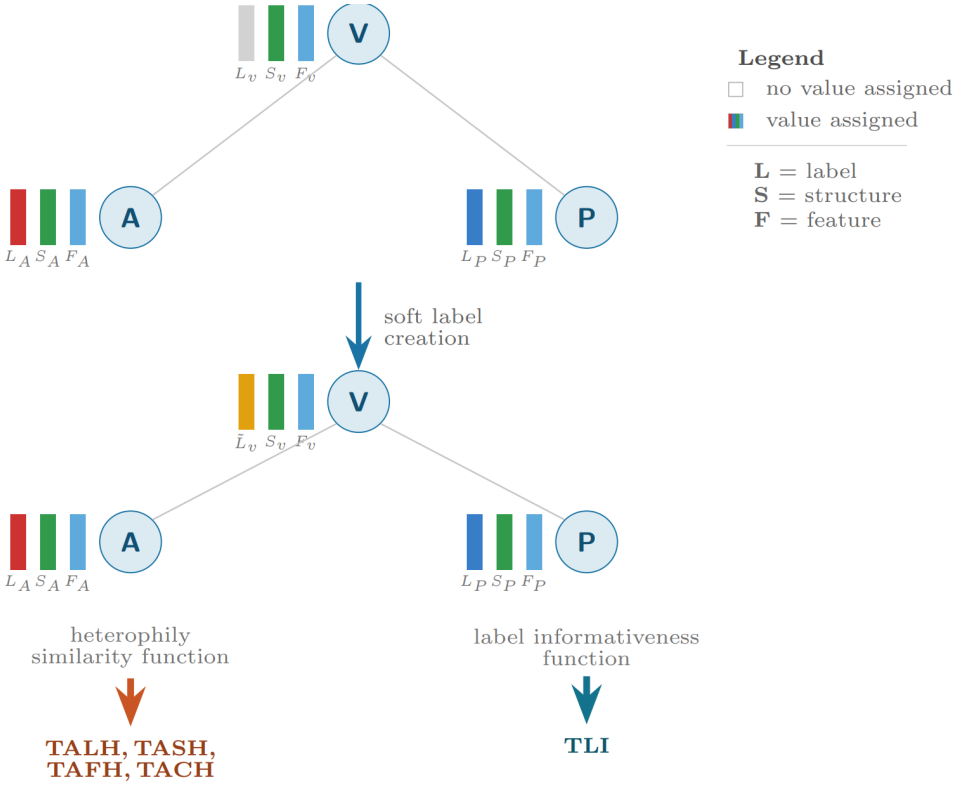


Figure 2: Proposed heterophily metrics construction framework

3.2 Component II (Q2): STAGE — Stiefel-Manifold Geometric Encoder

Gap. Existing HGNNs use unconstrained type projections, allowing semantically distinct node types to collapse in the shared latent space. No method provides a training-independent inter-type separation guarantee or jointly disentangles label-relevant from label-independent features.

Proposed Approach. Each node's features are first decomposed into label-relevant and label-independent components. Each node type is then assigned a dedicated subspace on a shared hypersphere, with boundaries enforced throughout training via Riemannian optimisation on the *Stiefel manifold*, yielding an unconditional inter-type angular separation guarantee **(G1)**. Within each subspace, class-specific anchor directions (initialised via the Tammes problem) are learned via a vMF variational objective, so nodes of the same type but different classes occupy geometrically distinct regions **(G2)**. STAGE is pre-trained once and plugged into any downstream HGNN as a drop-in initialisation module as shown in Figure 3.

Key novel claim: First unconditional algebraic guarantee of inter-type angular separation (G1) and structured intra-type class geometry (G2) in heterogeneous graph learning.

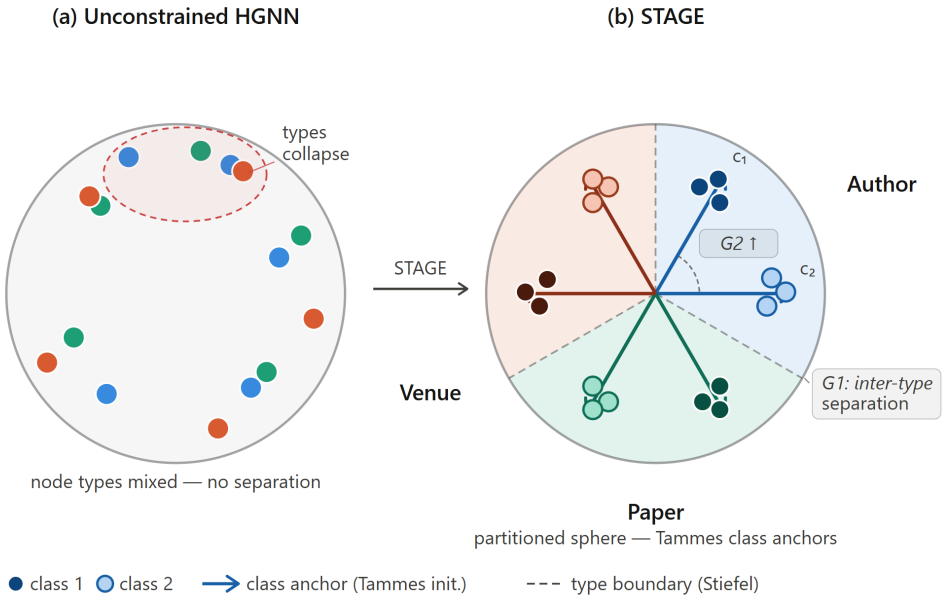


Figure 3: STAGE embedding framework

3.3 Component III (Q3): H²S-NN — Sheaf-Theoretic Static HGNN

Gap. All existing heterophily-aware HGNNs reduce the graph to meta-path-projected views, discarding direct cross-type edges. No model jointly handles inter/intra-type heterophily, meta-path higher-order relations, and over-smoothing within a unified framework.

Proposed Approach. The heterogeneous graph is *lifted* to a **typed 2D cellular complex presheaf**: nodes are 0-cells, typed edges are 1-cells, and ordered meta-path instances become 2-cells. Directed sheaf restriction maps from 0-cells to 1-cells are learned via an MLP over source node, target node, and edge embeddings; directed sheaf restriction maps from 1-cells to 2-cells are learned via an MLP over source node, target node, meta-path, type and sequence position. Neural diffusion minimises a three-level Hodge–Dirichlet energy. Based on sheaf theory, at convergence, the ODE solver yields harmonic node and edge cochains producing representations that are *globally coherent across three levels* (nodes, edges, and meta-path instances) through learned relation-specific restriction maps, rather than forcing node values to agree directly. Two heterophilic nodes may therefore retain discriminative representations while still being globally consistent with the relational and higher-order semantic structure of the graph, a property that flat node-level methods cannot achieve by design. The harmonic node embeddings are passed to an MLP classifier, trained end-to-end with cross-entropy loss, and the full pipeline is optimised iteratively as shown in Figure 4.

Key novel claim: First HGNN to jointly model nodes, edges, and meta-path instances in a single topological framework, provably mitigating over-smoothing under heterophily.

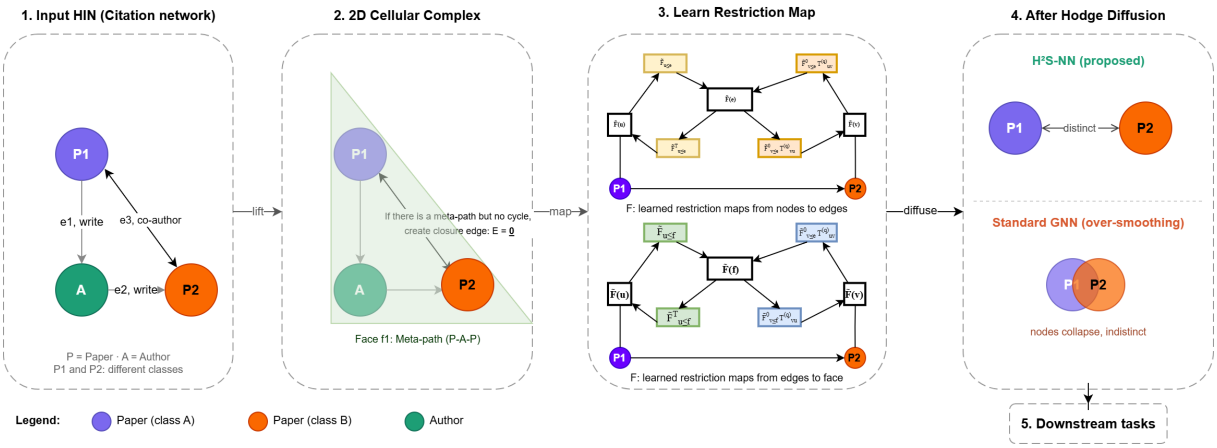


Figure 4: H²S-NN framework

3.4 Component IV (Q4): DistH^{2}SNN — Discrete-Time Dynamic Extension

Gap. All DT-DHGNNs conflate *persistent* (stable across snapshots) and *cumulative* (building over time) neighbourhood signals. A category-theoretic result [22] proves this causes irreversible information loss. No method detects or exploits heterophily drift across snapshots.

Proposed Approach. At each snapshot G_t, H^{2}S-NN produces Hodge embeddings. The neighbourhood is then *explicitly decomposed* into a persistent stream (edges surviving from G_{t-1}) and a cumulative stream (all observed edges in current and previous snapshots), with snapshot-gap time encoding $\Phi(\Delta t)$ for recency. A **temporal heterophily gate** computes the drift in per-relation heterophily scores between snapshots and injects this signal into two dedicated Mamba encoders, one per stream, whose hidden states are adaptively fused as shown in Figure 5.

Key novel claim: First DT-DHGNN with theoretically justified persistent/cumulative decomposition and explicit heterophily drift detection.

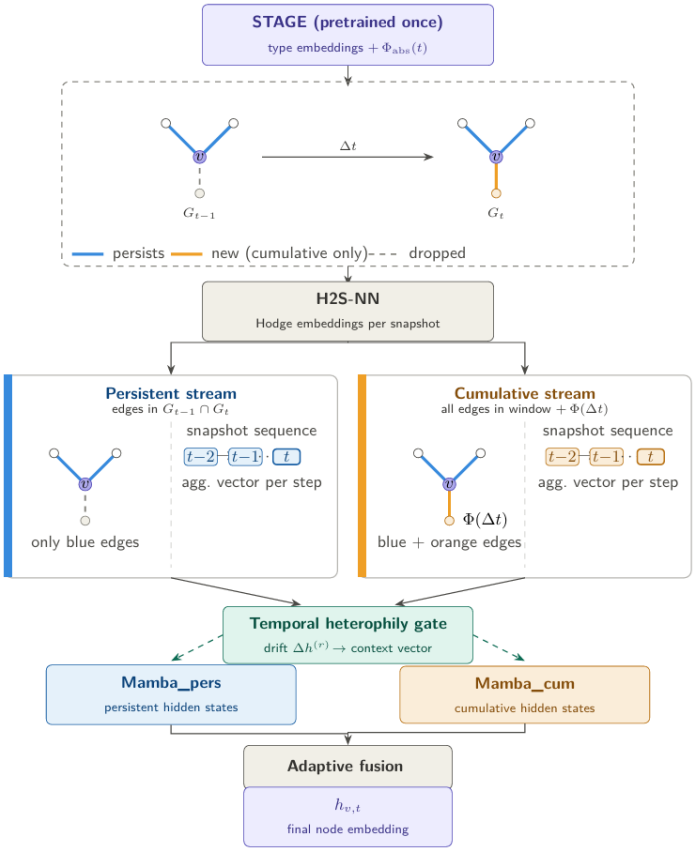


Figure 5: DistH^{2}SNN framework

3.5 Component V (Q5): ContH^{2}S-NN — Continuous-Time Extension

Gap. Existing continuous-time heterogeneous GNNs [23–26] do not account for heterophily. Irregular UNIX-timestamp event streams are incompatible with snapshot-based heterophily modeling.

Proposed Approach. ContH^{2}S-NN introduces two coupled streams. A non-dissipative anti-symmetric ODE maintains a global state per node across all events, preserving long-range history without information loss. At query time, a local *temporal complex* is constructed around the query pair, where 0-cells are node-time pairs (u, t') rather than nodes, 1-cells include both typed interaction edges and *lifeline edges* connecting consecutive participations of the same node, and 2-cells encode both persistent relational patterns and temporal meta-path instances. The existing Hodge diffusion from H^{2}S-NN runs on this local complex to produce heterophily-calibrated spatial representations. The two streams are coupled through a spatial buffer under gradient isolation, enabling joint learning while preserving the convergence guarantees of the Hodge ODE as shown in Figure 6.

Key novel claim: First continuous-time HGNN to model heterophily in a fully topological framework while keeping temporal neighborhood relations.

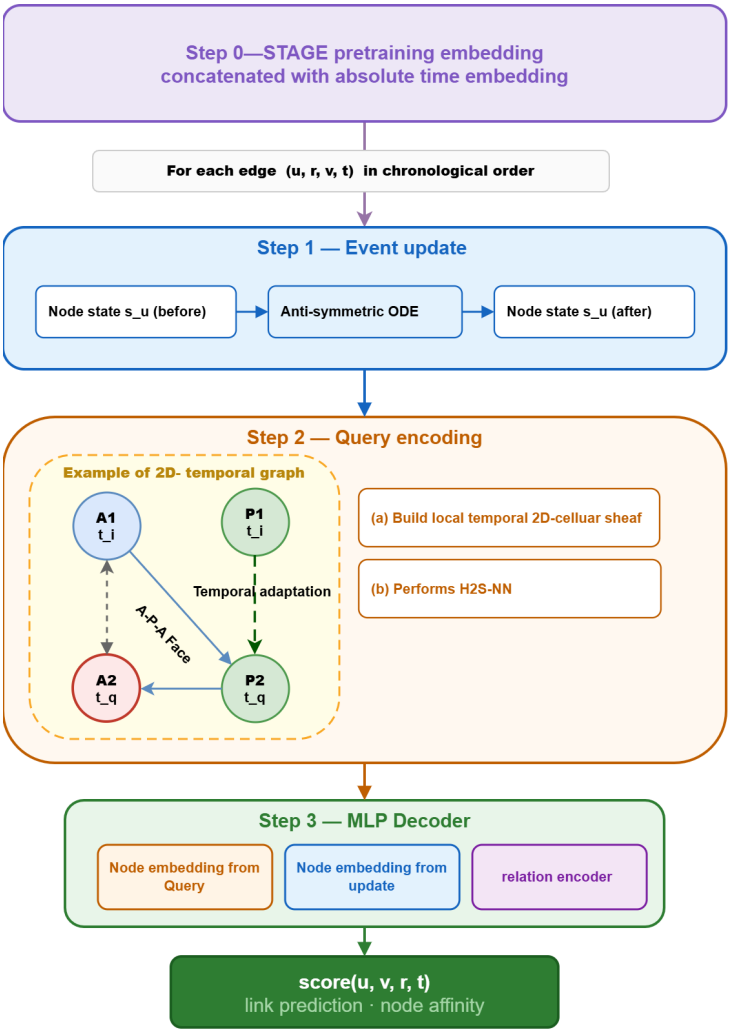


Figure 6: ContH^{2}S-NN framework

4 Study Programme in Murata's Lab

4.1 Ultimate Research Goal

My goal is a complete, theoretically grounded framework for learning on heterogeneous graphs under heterophily, spanning static, discrete-time, and continuous-time settings, resulting in four peer-reviewed publications over five years.

4.2 Year-by-Year Research Plan

Year / Phase	Research Activities and Milestones
Year 1 (M1)	**Foundation & Heterophily metrics & H^2S-NN.** Literature survey on sheaf-based GNNs, heterophilic GNNs, heterogeneous GNNs, topological deep learning, and neural diffusion models. Background study in category theory, sheaf theory, and algebraic topology as required for Q1 and Q3. Implement and finalise proofs for all five metrics (Q1). Implement full H^2S-NN (Q3). Comprehensive node classification experiments (Macro/Micro-F1). *Submit Paper 1 (H^2S-NN) to high-tier venue.*
Year 2 (M2)	**STAGE & Master's completion.** Literature survey on heterophilic GNNs, heterogeneous GNNs, representation learning, graph embedding, and geometric deep learning. Background study in differential geometry and Riemannian geometry as required for Q2. Implement and validate STAGE (Q2) on DBLP, IMDB, ACM, Freebase (node classification and clustering); ablation studies with t-SNE analysis. *Submit Paper 2 (Metrics + STAGE) to high-tier venue.* Complete Master's dissertation.
Year 3 (D1)	**Discrete-time Dynamic H^2S Extension & first doctoral presentation.** Literature survey on discrete-time dynamic heterogeneous GNNs and state-space sequence models. Implement DistH^2SNN: snapshot-gap encoding, temporal heterophily gate, dual Mamba encoders (Q4). Evaluate DistH^2SNN on Aminer, Ecomm, and DBLP (transductive link prediction, node affinity prediction), comparing against state-of-the-art baselines. *Submit Paper 3 (DistH^2SNN) to high-tier venue.* First doctoral interim presentation.
Year 4 (D2)	**Continuous-time Dynamic H^2S Extension.** Literature survey on continuous-time dynamic graph learning and neural ODE methods. Design and implement ContH^2S-NN: temporal edges, absolute/relative time encoding, and non-dissipative sheaf ODE (Q5). Evaluate on TGB 2.0 [31] datasets. *Submit Paper 4 (ContH^2S-NN) to high-tier venue.* Second doctoral interim presentation.
Year 5 (D3)	**Thesis completion & dissemination.** Integrate all four papers into a unified doctoral thesis with consistent notation and cross-paper theoretical connections. Complete pre-examination and final defence. Potential extensions: robust learning against structural/label/feature noise; automatic meta-path architecture search.

I sincerely hope that my application will be carefully reviewed, and I look forward to the opportunity to pursue the five-year integrated doctoral programme at the Murata Laboratory, Department of Computer Science, School of Computing, Institute of Science Tokyo (IGP(C) Fall Intake).

5 Related knowledge

5.1 Meta-paths

A *meta-path* is a schema-level path in a heterogeneous graph, defined by a sequence of node and edge types. It represents a typed relational pattern between specific node types and are widely used to capture higher-order semantic relations in heterogeneous graphs as shown in Figure 7.

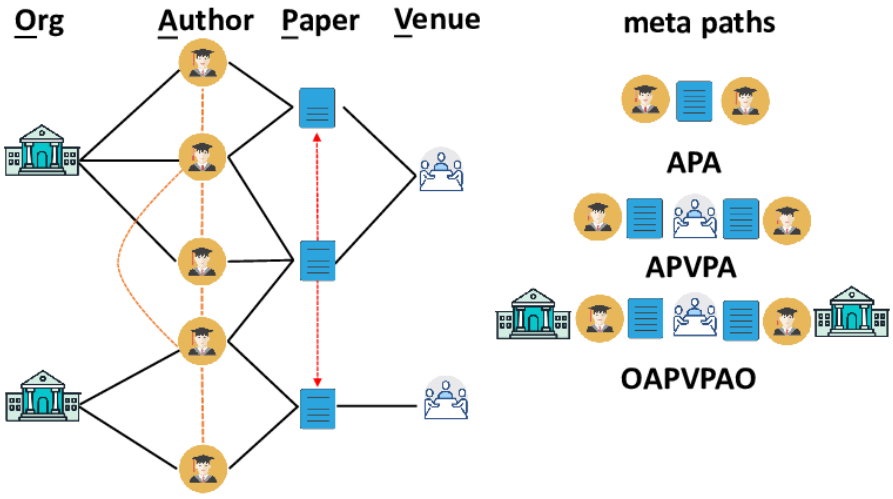


Figure 7: An illustrative example of a meta-path in an academic network. [32]

5.2 Heterogeneity and heterophily

As illustrated in Figure 8, Heterogeneity refers to the presence of multiple node and edge types (e.g., papers, authors, conferences), forming typed interactions. Heterophily describes whether connected nodes tend to have different labels or attributes. In heterogeneous graphs, models must capture type-specific relations, while under heterophily, naive neighborhood aggregation may mix dissimilar signals. Therefore, effective HGNNs should jointly account for both.

5.3 Discrete- and Continuous-Time Dynamic Graph

As shown in Figure 9, dynamic graphs can be modeled in two ways. In the *discrete-time* setting, the graph is represented as a sequence of snapshots, where each snapshot summarizes the graph during a time interval. This setting is suitable when data are grouped by periods such as days or months. In the *continuous-time* setting, the graph is modeled as a stream of timestamped events, so interactions are processed at the exact time they occur. This is suitable when events happen irregularly [33–35].

5.4 Over-smoothing

is the phenomenon where, as network depth (number of layers) increases, node embeddings become too similar and eventually indistinguishable. This homogenization destroys node-specific features, leading to performance degradation in tasks like node classification, where distinct representations are necessary.

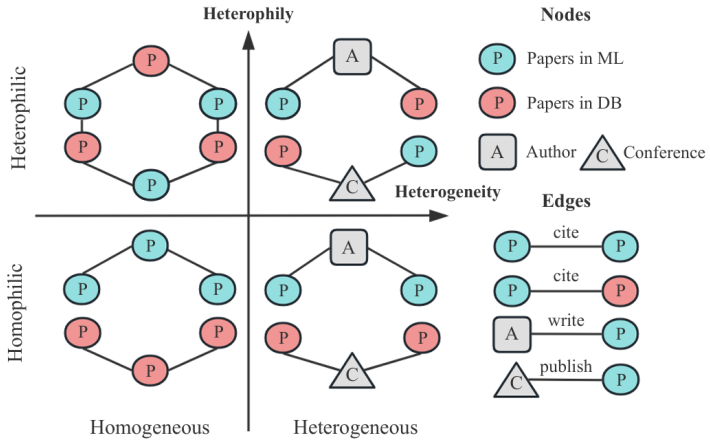
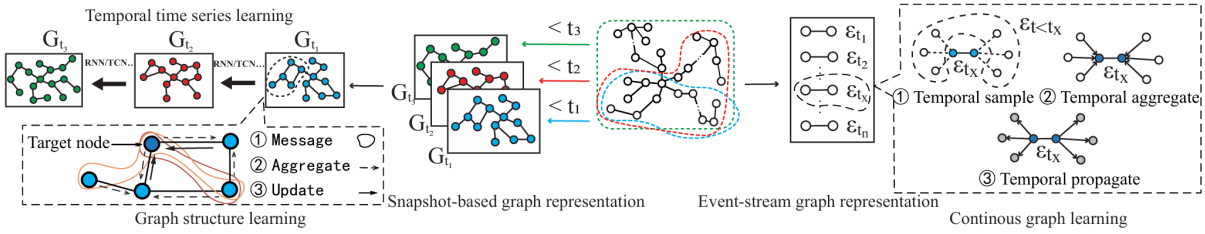


Figure 8: Example of four types of graphs based on heterogeneity and heterophily. The horizontal axis represents heterogeneity, divided into homogeneous and heterogeneous graphs from left to right. The vertical axis represents heterophily, divided into homophilic and heterophilic graphs from bottom to top. "Paper" denotes the target node to be classified. [17]



(a) Discrete-time dynamic graph (b) Continous-time dynamic graph

Figure 9: A toy example of dynamic graph representation and learning [36].

5.5 Homophilic metrics desired properties

Property	Description
Maximal Agreement (Max)	Score equals +1 if and only if all edges have perfect atom agreement
Minimal Agreement (Min)	Score equals −1 if and only if all edges have zero atom agreement
Asymptotic Constant Baseline (ACB)	Score concentrates at 0 on graphs drawn from the typed null model
Empty Class Tolerance (ECT)	Adding an empty class leaves the score unchanged
Monotonicity (Mon)	Adding an edge between two nodes of the same class increases the score, while adding an edge between two nodes of different classes decreases it

Table 3: Five axiomatic desiderata for heterophily metrics.

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